

Astropy and MCMC exercise

1 Coordinate Systems

There are different coordinate systems to locate astronomical objects in the sky. The most common ones are the celestial (equatorial) coordinate system, which uses Right Ascension (RA) and Declination (Dec), and the galactic coordinate system, which uses galactic longitude (l) and galactic latitude (b).

1.1 Astronomical bodies in celestial coordinates

The coordinates of the following astronomical bodies are obtained from Wikipedia (links at the end of the document):

- Large Magellanic Cloud (LMC)
- Sagittarius A* (SagA)
- Andromeda Galaxy (M31)
- Sirius (Alpha Canis Majoris)
- Pleiades (M45)

and two functions are created to transform their celestial coordinates from hours, minutes, seconds (for Right Ascension) and degrees, arcminutes, arcseconds (for Declination) to decimal degrees:

```
# Function to convert from h,m,s to degrees
def hms_to_deg(h, m, s):
    return 15 * (h + m / 60 + s / 3600)

# Function to convert from d,m,s to degrees
def dms_to_deg(d, m, s):
    sign = 1 if d >= 0 else -1
    return sign * (abs(d) + m / 60 + s / 3600)
```

Then, these coordinates are saved as `astropy.coordinates.SkyCoord` and printed:

```
LMC coordinates: <SkyCoord (ICRS): (ra, dec) in deg
(80.89166667, -69.75666667)>
SagA coordinates: <SkyCoord (ICRS): (ra, dec) in deg
(266.41681667, -30.99217503)>
Andromeda coordinates: <SkyCoord (ICRS): (ra, dec) in deg
(10.68470833, 41.26875)>
Sirius coordinates: <SkyCoord (ICRS): (ra, dec) in deg
(101.28715542, -17.28388414)>
M45 coordinates: <SkyCoord (ICRS): (ra, dec) in deg
(56.75, 24.1167)>
```

1.2 Coordinate transformation into galactical coordinates

To continue with the exercise, the celestial coordinates of the astronomical bodies are transformed into galactic coordinates using the `.galactic` attribute of the `SkyCoord` objects.

```
LMC Galactic coordinates: <SkyCoord (Galactic): (l, b) in deg
(280.46606141, -32.88914978)>
SagA Galactic coordinates: <SkyCoord (Galactic): (l, b) in deg
(358.25020354, -1.0797243)>
Andromeda Galactic coordinates: <SkyCoord (Galactic): (l, b) in deg
(121.17432906, -21.5733088)>
Sirius Galactic coordinates: <SkyCoord (Galactic): (l, b) in deg
(227.74811911, -9.13685218)>
M45 Galactic coordinates: <SkyCoord (Galactic): (l, b) in deg
(166.57063837, -23.52114314)>
```

1.3 Aitoff projection for both coordinate systems

The celestial and galactic coordinates of the astronomical bodies are then plotted using an Aitoff projection, which is a type of map projection commonly used in astronomy to represent the entire sky on a flat surface.

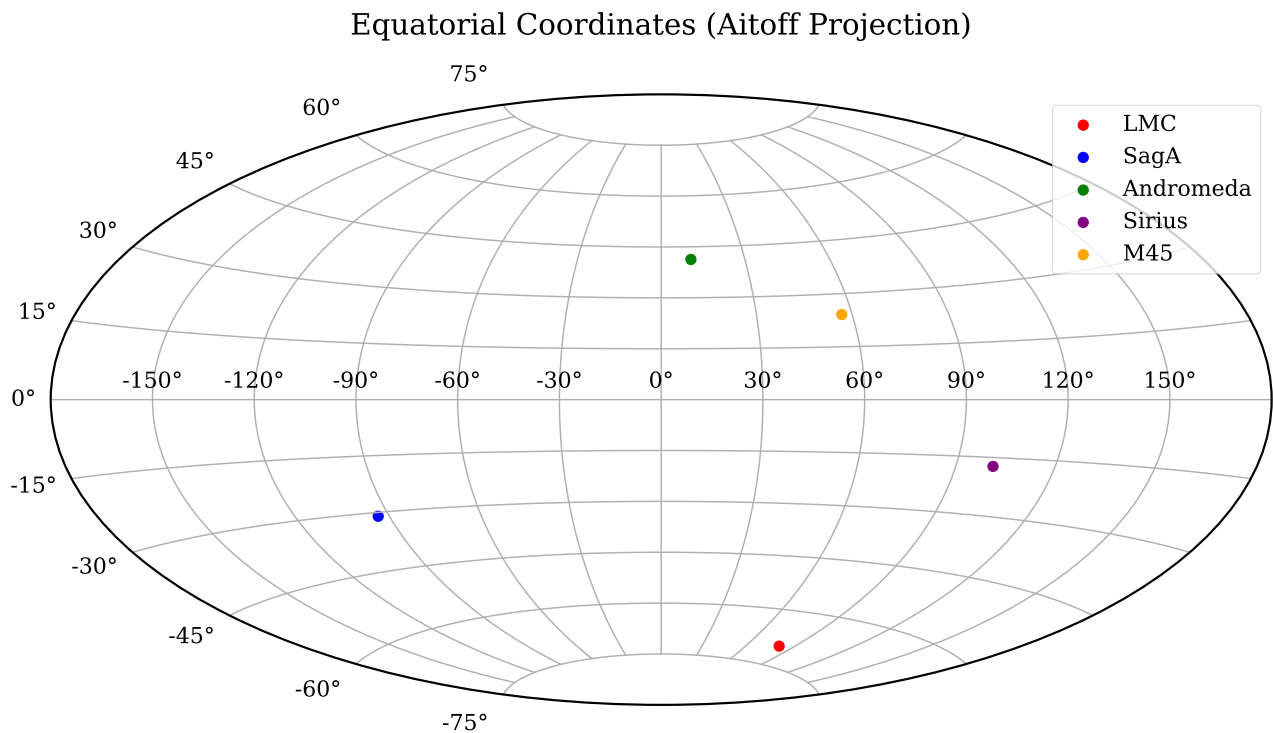


Figure 1: Aitoff projection of celestial coordinates (RA, Dec) of the astronomical bodies.

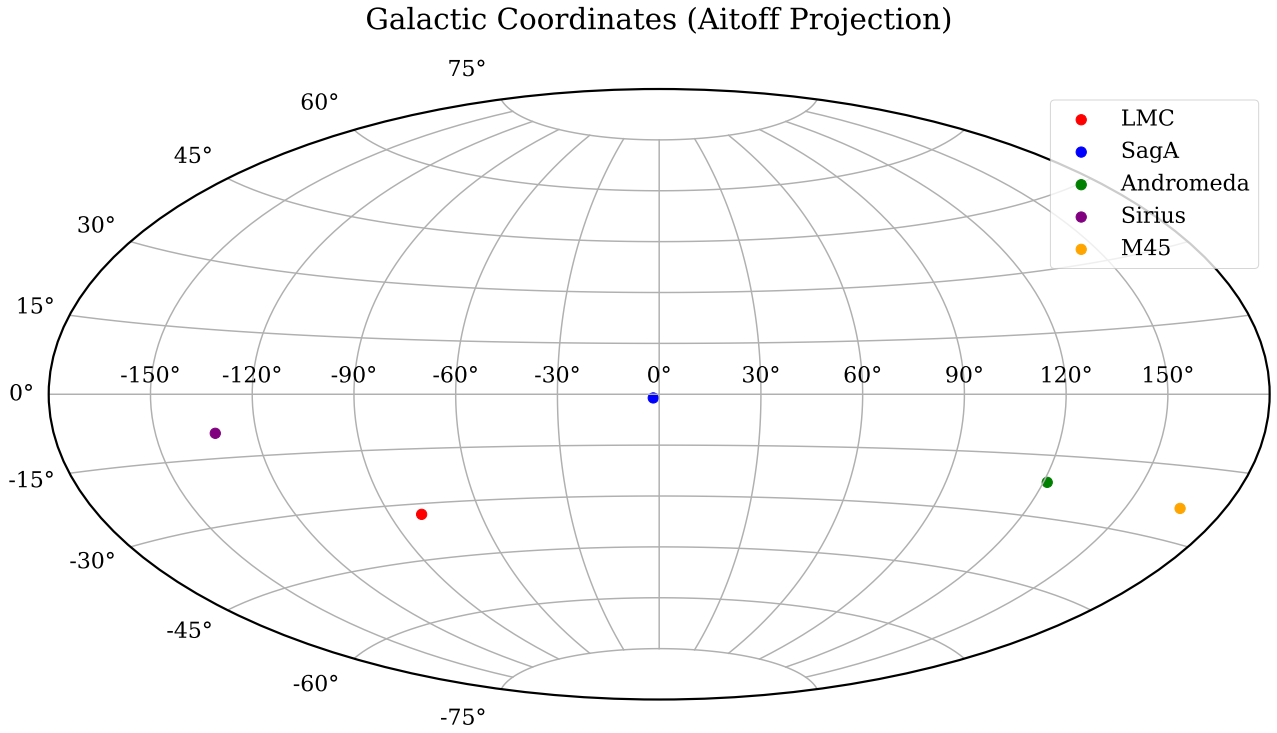


Figure 2: Aitoff projection of galactic coordinates (l, b) of the astronomical bodies.

2 Super Novae Cosmology

Super Novae are useful standard candles in cosmology because of the uniformity of their light curves. These allow us to measure distances in the universe and thus constrain cosmological parameters. In this exercise the Planck18 model is used.

2.1 Distance modulus for super novae

The distance modulus μ is defined as the difference between the apparent magnitude m and the absolute magnitude M of an astronomical object. However, there is an alternative expression to calculate it:

$$\mu = 5 \times \log_{10}(D_L[10 \text{ pc}]), \quad (1)$$

where D_L is the cosmological luminosity distance, defined as:

$$D_L = (1 + z)D_M = (1 + z) \frac{c}{H_0} \int_0^z \frac{1}{E(z')} dz', \quad (2)$$

where D_M is the transverse comoving distance, z is the redshift, c is the speed of light, H_0 is the Hubble constant, and $E(z)$ is the dimensionless Hubble parameter.

The Planck18 cosmology model is used to compute some relevant cosmological quantities with the following code:

```
from astropy.cosmology import Planck18

cosmo = Planck18

print(cosmo)
print(cosmo.h)
print(cosmo.hubble_distance)
```

```
print(cosmo.hubble_time)
print(cosmo.comoving_distance(0.5)) # the comoving distance to z=0.5
print(cosmo.comoving_distance(1).to(u.lightsecond)) # the comoving
distance to z=1 in lightseconds
```

and the output is shown below:

```
FlatLambdaCDM(name="Planck18", H0=67.66 km / (Mpc s), Om0=0.30966,
  Tcmb0=2.7255 K, Neff=3.046, m_nu=[0. 0. 0.06] eV, Ob0=0.04897)
0.6766
4430.866952409105 Mpc
14.451555153425794 Gyr
1946.4166347871185 Mpc
3.4950289385104154e+17 lsec
```

2.2 Distance modulus Vs. Redshift

A raw data file containing super novae observations is provided. The data contains the redshift and distance modulus for each super nova, along with a probability indicating the likelihood of the event being a super nova. The data is filtered to include only those events with a high probability (greater than 0.9) of being true super novae. Using this filtered data, the distance modulus is plotted against redshift and compared with the theoretical predictions from the Planck18 model.

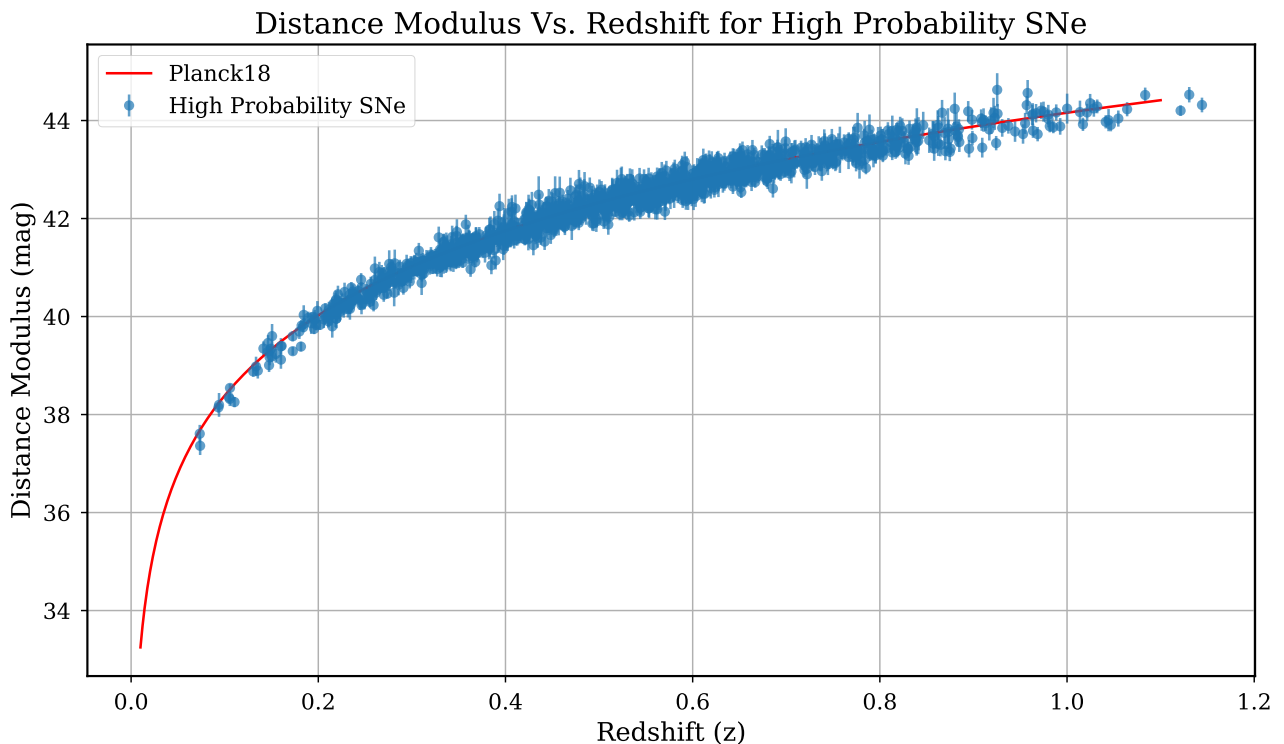


Figure 3: Distance Modulus Vs. Redshift using the data from the file specified in the notebook for events with high probability to be super novae.

The behavior of the distance modulus as a function of redshift is consistent with the expectations from the Planck18 cosmological model, indicating that the filtered super novae data aligns well with the theoretical predictions.

2.3 Two parameter MCMC

A two parameter MCMC is also performed by sampling H_0 and Ω_m for a flat Λ CDM cosmology to estimate the cosmological parameters Ω_Λ and ω_b , like in the example of the notebook.

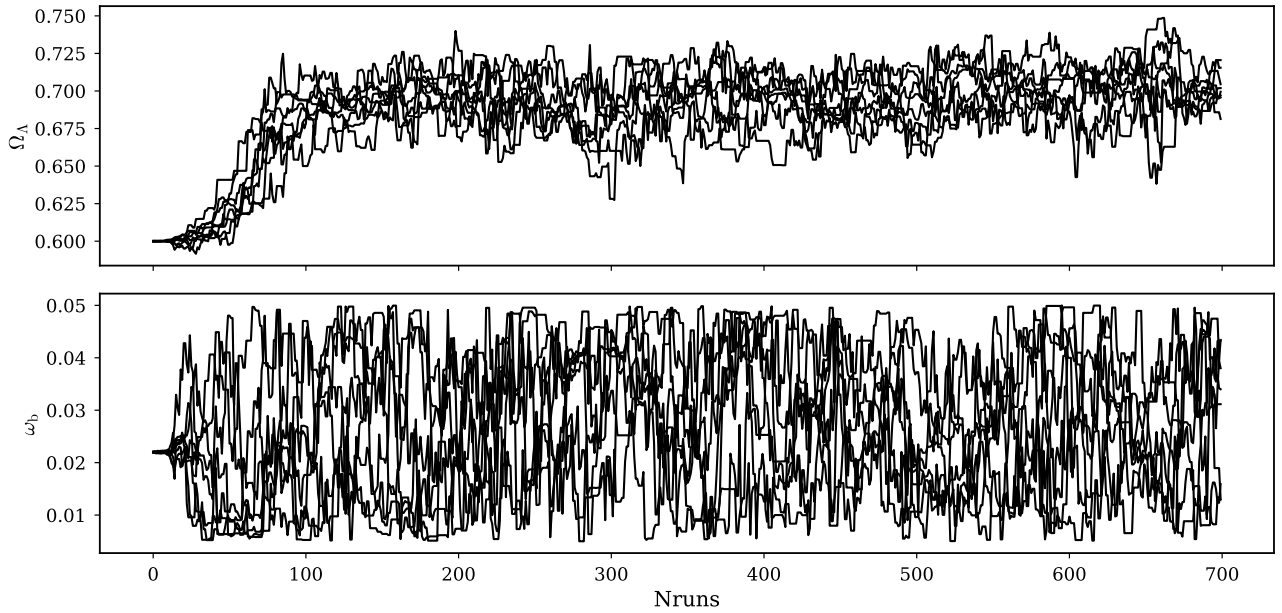


Figure 4: Trace plots for the MCMC sampling of the cosmological parameters Ω_Λ and ω_b for 700 steps with 8 walkers

The first 100 steps of the MCMC chains are discarded as burn-in to ensure that the samples used for analysis are drawn from the stationary distribution. Then, a corner plot is generated to visualize the posterior distributions of the sampled parameters in Figure 5.

2.4 Compute from the cut samples the mean and 68% confidence interval

Finally, the mean values and 68% confidence intervals for the cosmological parameters Ω_Λ and ω_b are computed from the MCMC samples.

Parameter	Mean	68% Confidence Interval
Ω_Λ	0.695	[0.679, 0.712]
ω_b	0.028	[0.013, 0.042]

Table 1: Mean and 68% confidence intervals for the cosmological parameters Ω_Λ and ω_b .

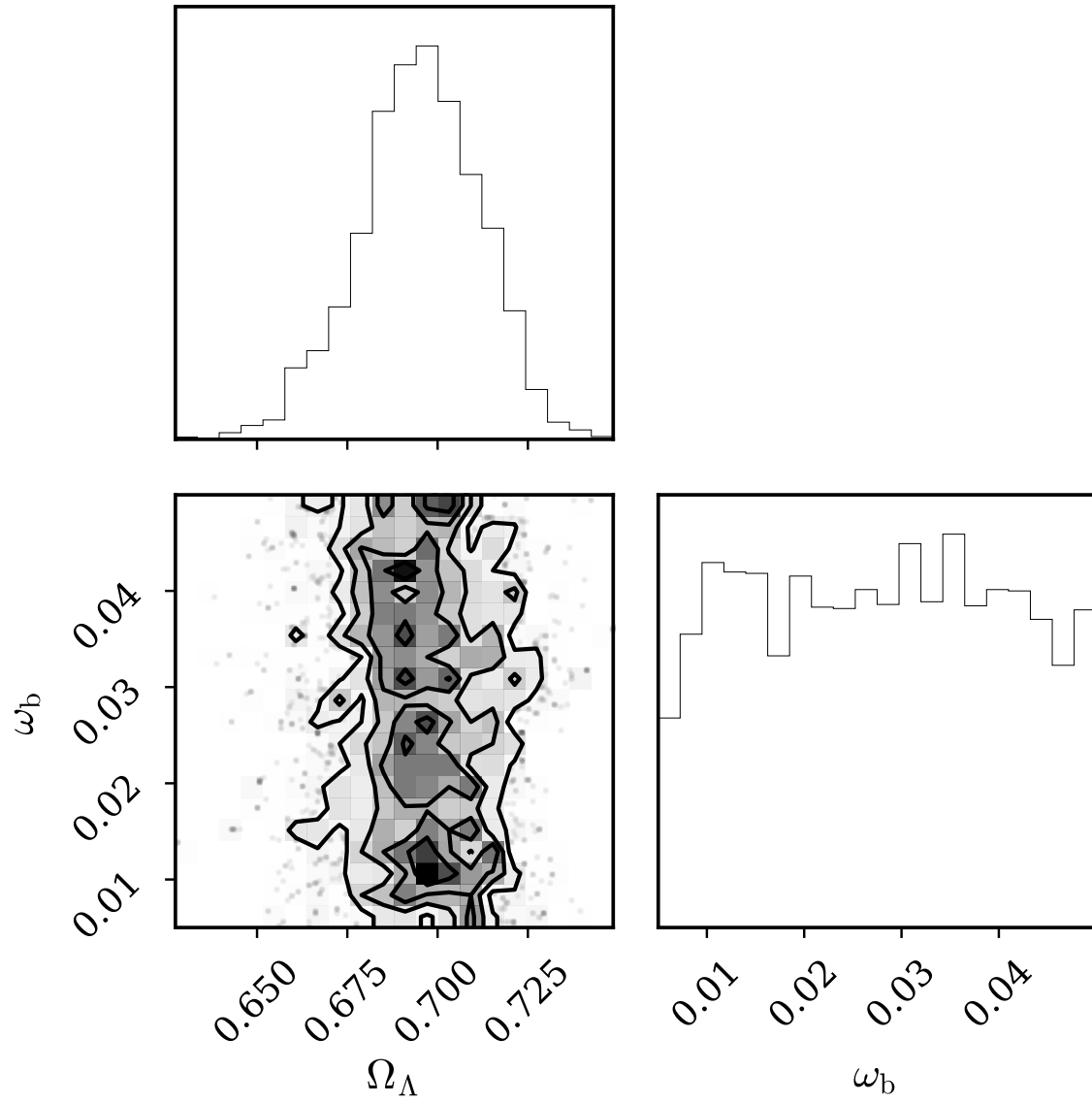


Figure 5: Corner plot of the posterior distributions of the cosmological parameters Ω_Λ and ω_b .

Wikipedia links to get the coordinates

https://en.wikipedia.org/wiki/Large_Magellanic_Cloud
https://ca.wikipedia.org/wiki/Sagitari_A*
https://ca.wikipedia.org/wiki/GalÃxia_d%27AndrÃsmeda
<https://ca.wikipedia.org/wiki/SÃnrius>
[https://es.wikipedia.org/wiki/PlÃyades_\(astronomÃa\)](https://es.wikipedia.org/wiki/PlÃyades_(astronomÃa))